Nephrotic syndrome

https://doi.org/10.1038/s41581-024-00873-7

Anti-nephrin autoantibodies: a paradigm shift in podocytopathies

Zhao Cui & Ming-hui Zhao

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A new study demonstrates that anti-nephrin autoantibodies are not merely markers but also actively contribute to the pathogenesis of minimal change disease and primary focal segmental glomerulosclerosis. This insight not only provides a non-invasive diagnostic alternative to kidney biopsies, but also suggests potential for novel targeted therapies.

Refers to Hengel, F. E. et al. Autoantibodies targeting nephrin in podocytopathies. *N. Engl. J. Med.* https://doi.org/10.1056/NEJMoa2314471 (2024).

Patients with primary nephrotic syndrome — minimal change disease (MCD) and primary focal segmental glomerulosclerosis (FSGS) in adults, idiopathic nephrotic syndrome in children — often respond to immunosuppressive treatment, including B cell targeted therapy[1]. Following earlier studies[2,3], Hengel and colleagues[4] now provide compelling evidence that anti-nephrin autoantibodies have a substantial role in podocytopathies, specifically in MCD and primary FSGS. These findings could redefine our understanding and approach to the management of these glomerular diseases.

In this study[4], the researchers developed a hybrid assay that combined IgG immunoprecipitation with a nephrin ELISA, allowing them to detect and quantify circulating anti-nephrin antibodies. They used this assay to assess the prevalence of anti-nephrin autoantibodies across different glomerular diseases in a cohort that comprised 539 patients (357 adults and 182 children) and 117 healthy individuals. Anti-nephrin autoantibodies were detected in 44% of adults with MCD and in 9% of those with primary FSGS. Notably, these autoantibodies were present in 52% of children with idiopathic nephrotic syndrome. The autoantibody prevalence rose, respectively, to 69% and 90% among patients with MCD or idiopathic nephrotic syndrome who had active disease and were not receiving immunosuppression. Longitudinal analyses demonstrated a correlation between anti-nephrin autoantibody levels and disease activity, suggesting their potential as biomarkers. Of note, the low prevalence of these antibodies in primary FSGS suggests that the underlying immune-mediated mechanisms have not been fully elucidated. Not only might more target antigens be discovered, but the autoantibodies involved might not be limited to the IgG class and might also include IgM, considering the IgM and C3 deposits observed on segmental sclerotic lesions and their close associations with the severity of nephrotic syndrome, unfavourable treatment responses and worse kidney outcomes[5].

The researchers used an experimental mouse model[4] to elucidate the pathogenic mechanisms associated with anti-nephrin antibodies. Immunization of mice with murine nephrin triggered autoantibody development and induced a nephrotic syndrome resembling MCD, with diffuse effacement of podocyte foot processes. The localization of IgG at the slit diaphragm and its partial colocalization with nephrin indicated antigen–antibody binding. This binding increased phosphorylation of nephrin at tyrosine residue Y1191 (corresponding to Y1176 in human nephrin), which led to cytoskeletal reorganization. Proteomic data further suggested an increase in nephrin endocytosis and disruption of the slit diaphragm. Enhanced endocytosis might explain the lack of detectable IgG and electron-dense deposits in patients with MCD, and the clinical course of abrupt onset and rapid remission after treatment with corticosteroids might also reflect precipitous changes in these intracellular processes.

These new data[4] align with previous findings emphasizing the importance of kidney-specific autoantibodies in the pathology of autoimmune glomerular disease (Fig. 1). For example, patients with anti-glomerular basement membrane (GBM) disease develop autoantibodies against the non-collagenous domain 1 of the α3 chain of type IV collagen on the GBM[6]. The IgG-degrading enzyme of *Streptococcus pyogenes* (IdeS) can be beneficial in managing this disease[7]. For primary membranous nephropathy, the discovery of phospholipase A2 receptor (PLA2R) as the autoantibody target antigen on podocytes in ~70% of patients changed understanding of the disease and prompted revision of diagnostic and therapeutic algorithms[8]. In these patients, B cell-depleting agents were superior to cyclosporin in achieving both clinical and immunological remission[9]. In the context of MCD and FSGS, the identification of anti-nephrin autoantibodies suggests similar pathogenic mechanisms to those of other autoantibody-mediated glomerular diseases and explains the efficacy of CD20 monoclonal antibodies[1].

Downstream of autoantibody binding, the mechanisms driving kidney injury can vary substantially, which underlies the diverse clinical presentations observed, ranging from mild proteinuria to rapidly progressive glomerulonephritis. These mechanisms might include inflammatory effects of IgG subclasses via complement activation and antibody-dependent cell-mediated cytotoxicity. In addition, intracellular alterations in podocytes, including the quickly reversible nephrin phosphorylation and actin assembly observed in MCD, or the relatively slow regeneration of podocyte structural proteins (synaptopodin and nephrin-like protein 1) following their proteolytic degradation in membranous nephropathy[10] can disrupt the glomerular filtration barrier. Finally, the location of antigens (for example, on podocytes or the extracellular matrix) might also affect the disease course. Clarifying the mechanisms driving kidney injury downstream of antigen–antibody binding might lead to novel therapeutics, beyond the direct targeting of autoantibodies.

Rather than being distinct entities, MCD and primary FSGS seem to represent a spectrum of podocyte injury, as evidenced by their shared causes, including drugs, infections, genetic mutations and autoantibodies. Accordingly, a more accurate and practical classification system for

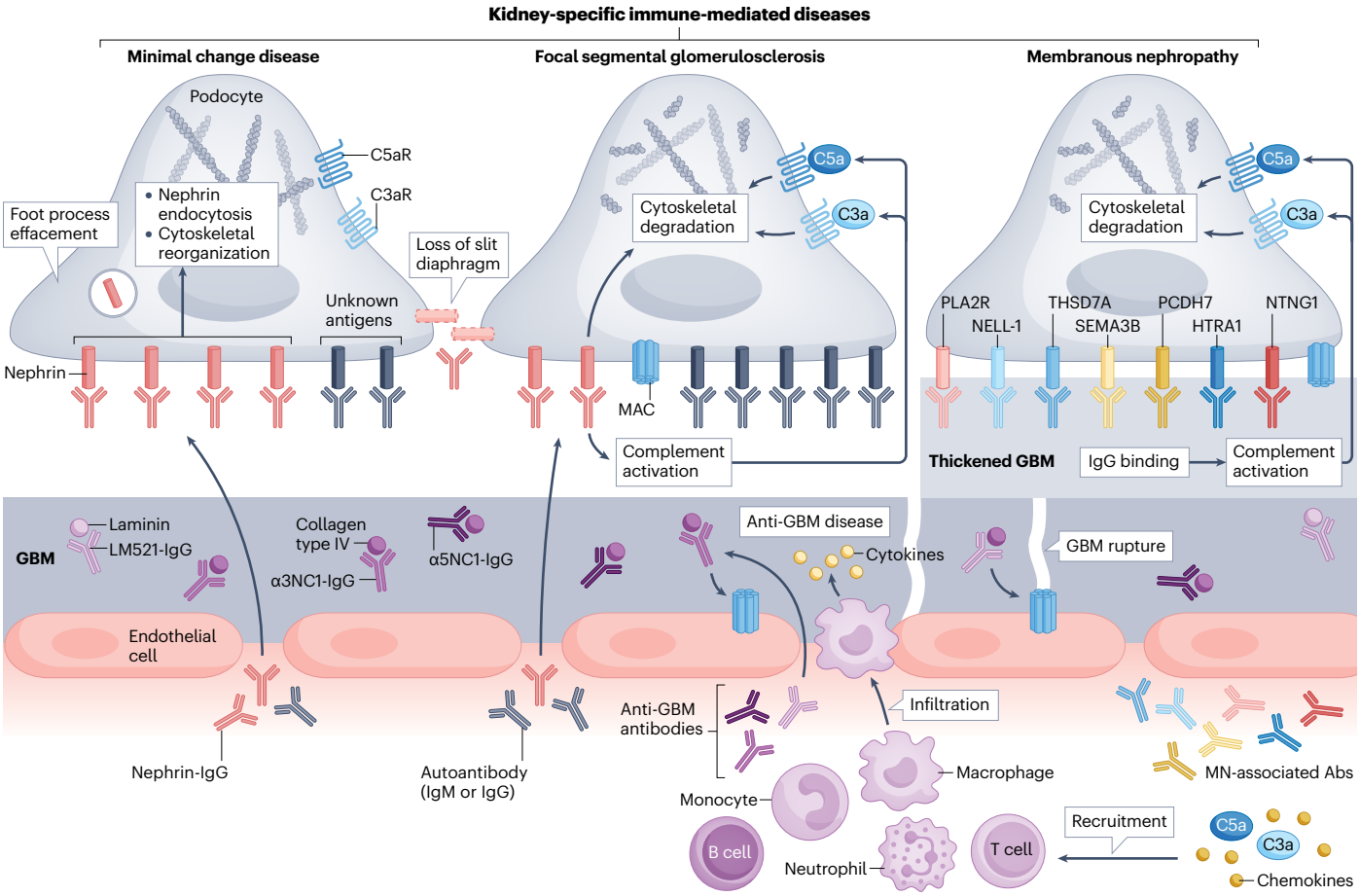


Fig. 1 | Kidney-specific immune-mediated glomerulonephritis. Autoantibodies targeting kidney-specific antigens have key pathogenic roles in podocytopathies, and their identification has important diagnostic and therapeutic implications. Kidney autoantigens include the phospholipase A2 receptor (PLA2R) on podocytes (implicated in membranous nephropathy (MN)) and glomerular basement membrane (GBM) antigens such as the collagen type IV domains α3NC1 and α5NC1, and laminin 521 (LM521; implicated in anti-GBM disease). A new study reports that in most cases of minimal change disease (MCD) and in a small proportion of primary focal segmental glomerulosclerosis (FSGS) cases, anti-nephrin autoantibodies are present in circulation and bind to nephrin located on the slit diaphragm between podocytes. This binding leads to nephrin endocytosis and cytoskeleton reorganization, which leads to nephrotic syndrome. Downstream of autoantibody binding in the kidney, the mechanisms of cell injury and tissue damage vary substantially, depending on the inflammatory characteristics of the antibodies and the alterations in the intracellular downstream pathways of the autoantigens involved. Abs, antibodies; C3aR, complement C3a receptor; NELL-1, neural epidermal growth factor-like protein 1; THSD7A, thrombospondin type-1 domain-containing 7 A; SEMA3B, semaphorin 3B; PCDH7, protocadherin 7; HTRA1, high-temperature requirement A serine peptidase 1; NTNG1, netrin G1; MAC, membrane attack complex.

podocytopathies, based on pathogenesis — including autoantibody detection — rather than histological description might aid diagnosis and long-term disease management. The newly reported hybrid immunoprecipitation–ELISA used to detect anti-nephrin antibodies also offers a non-invasive diagnostic alternative to kidney biopsies, which is particularly important for children with idiopathic nephrotic syndrome, for whom kidney biopsy is not routinely performed. A more convenient and accessible assay needs to be developed and validated in large cohorts for broader application in clinical practice.

Zhao Cui [1,2] **& Ming-hui Zhao** [1,2] ✉
[1]Renal Division, Peking University First Hospital, Beijing, China.
[2]Institute of Nephrology, Peking University, Beijing, China.
✉e-mail: mhzhao@bjmu.du.cn

Published online: 30 July 2024

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Acknowledgements

The authors thank Huang Kuang for creating the original artwork.

Competing interests

The authors declare no competing interests.